Ethnicity and Agricultural Attachment in the Post-WWII Southwest

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Abstract

World War II (WWII) was a large and multifaceted labor market shock that catalyzed improvements in the economic outcomes of Black men in the United States. Manufacturing booms, labor shortages, and federal efforts to lower employment barriers for underrepresented groups pulled men off the farm and into factory jobs, which led to persistent economic gains. The same can not necessarily be said for Hispanic men in the Southwest, of whom over 20 percent were employed in agriculture still in 1950. This paper investigates the drivers of Hispanic men's relative attachment to the farm sector in the Southwest during this pivotal period in history, with a particular focus on the role of introducing the Bracero program. Using linked census data from 1940 and 1950, I find that Hispanic men were more likely to *enter* the agricultural sector than Black men by 1–2 percentage points. I also show evidence consistent with two potential mechanisms: language entry barriers in the non-farm sector and increasing demand for Spanish-speaking farm workers. In ongoing work, I explicitly consider the role of the foreign workers in shaping the occupational choice of natives, and I explore whether this attachment to the farm sector played a role in the economic trajectory for Hispanics in the Southwest.

JEL codes: J15, J43, J61, N32, N52

Keywords: agricultural labor, Hispanic workers, occupational choice, World War II, Bracero Program, ethnicity

1 Introduction

It is well-documented that World War II (WWII) was a positive turning point for the economic outcomes of Black men in the United States (US). In metropolitan areas, wartime manufacturing expansions increased incomes and rates of skilled employment for Black men relative to White men and these gains were persistent for decades after the war concluded (Aizer et al., 2022). Even in the South, Black men experienced greater upward occupational mobility in areas with higher casualty rates of White soldiers previously employed in semi-skilled occupations (Ferrara, 2022), in spite of Jim Crow-era discrimination. Importantly, the gains Black men experienced during and after the war had positive intergenerational effects, with the next generation of Black children getting more education (Aizer et al., 2022). This expansion of economic opportunities for Black men coincided with a dramatic exit from agricultural occupations (Wolfbein, 1947). Despite the consensus that Black outcomes improved after WWII, there has been relatively little study of whether the same was true for Hispanic men. Over 70 percent of the national Hispanic population in 1940 lived in 5 states - California, Arizona, New Mexico, Colorado, and Texas (hereafter the Southwest) - and roughly 35 percent of both Black and Hispanic men in the Southwest worked in agriculture.[1] However, while the share of Black men employed in agriculture in these states fell to 14 percent by 1950, the magnitude of exit was far less pronounced for Hispanic men, such that 23 percent of Hispanic workers were still employed in agriculture by 1950.

In this paper, I ask why Hispanic men in the Southwest exhibit more attachment to farmwork than Black men in an era of unprecedented economic expansion in the non-farm sector. Using linked census data from 1940 and 1950, I find evidence that this pattern was primarily driven by differential exposure to new agricultural technology, language barriers, and increased demand for Spanish-speaking farm workers. Black farm workers were concentrated in regions specializing in field crops, primarily cotton and rice, while Hispanic farm workers were concentrated in regions producing specialty crops, and I find that this regional segmentation explains much the difference in rates of exit from the farm sector between these two groups. However, Hispanic men were also 1 to 2 percentage points more likely to *enter* farm work by 1950 even after accounting for regional variation. English language ability (ELA) was a strong predictor of entry for Hispanic men, such that an individual who spoke English with 50 percent probability was 5 percentage points more likely

[1] All employment-related statistics for Hispanics are based on the population of citizen males in the labor force. When including non-citizens, 38 percent of Hispanic men worked in agriculture.

to enter farm work than an individual who spoke English with near 100 percent probability, holding all else constant. I find that some of this effect is attributed to language barriers for entering the manufacutring sector, but that language barriers were somewhat mitigated in areas where Hispanics were proportionally represented in manufacturing prior to the war. Finally, I find some preliminary evidence that there was an increase in demand for Spanish-speaking farm workers. Among men who remained in farm work between the two census periods, Hispanics experienced income growth that was over 14 percent larger than that of Black and White incumbents, and nearly all of these gains were for men with low-expected ELA. This is consistent with domestic Hispanic workers being more complementary with foreign farm workers from Mexico that arrived after the introduction of the WWII Bracero Program.

This is the first paper that explores the relative attachment of Hispanic men to the farm sector during this critical period of history. This work contributes to the large body of economic research exploring the effect of WWII on economic outcomes for marginalized groups, which has primarily focused on Black men (Turner and Bound, 2003; Aizer et al., 2022; Ferrara, 2022; Lleras-Muney et al., 2025) and women (Goldin, 1991; Fernandez et al., 2004; Goldin and Olivetti, 2013). Existing work studying Hispanics during this period focused on the effect of military service during WWII (Fishback et al., 2024), so by primarily focusing on the homefront I further our understanding of the economic limitations Hispanic men faced during and after the war. With the war came the Bracero program, a series of three agreements between the US and Mexico that spanned more than two decades and facilitated the employment of over 4 million guest workers from Mexico on farms in the US. Despite the program's lasting impact of solidifying Mexican migrants as the backbone of the US farm labor force, relatively little is known about the impact of this program on domestic farm workers. Recently, Clemens et al. (2018) found that terminating the program in 1964 had no effect on farm wages, and instead accelerated the adoption of new labor-saving technology. My paper instead focuses on outcomes for men who were farm workers when the program was first introduced, and thus sheds new light on the differential effect that guest workers had on the employment decisions of farm workers. Finally, this paper contributes to the literature of determinants of occupational choice for minority groups. More targeted work has looked at occupational choices for Jews in Europe (Brenner and Kiefer, 1981; Botticini and Eckstein, 2005), Arabs in Israel (Kasir and Yashiv, 2020), and Blacks in the US (Haggerty and Johnson, 1995), while other papers have looked more broadly at ethnic minorities in developed countries like the United Kingdom (Battu et al., 2011), Norway (Abrahamsen and Drange, 2015), and Switzerland (Pregaldini et al., 2022). A handful of economics papers have studied occupational

choice and occupational segregation patterns for Hispanics in the US (Toussaint-Comeau, 2006; Hellerstein et al., 2009; Alonso-Villar et al., 2013), but to my knowledge most, if not all, of this work is embedded in more contemporary settings. My focus on the period following WWII provides important historical context to better understand the overrepresentation of Hispanics in agriculture that have persisted over time in the Southwest.

This paper proceeds as follows. In section 2, I briefly provide some historical context on the Southwest during the 1940s. Then in section 3, I describe my data sources and provide some stylized facts showcasing differences in agricultural attachment between Hispanic and Black men. Section 4 describes my baseline empirical approach. My main results are given in section 5 and I explore potential mechanisms in section 6. Section 7 concludes.

2 Historical Background

2.1 Hispanics in the Southwest

The Mexican-American War concluded in 1848 with the Treaty of Guadalupe Hidalgo, in which Mexico ceded over 500,000 square miles to the United States, land that ultimately became California, Nevada, Utah, Arizona, New Mexico, Colorado and Texas (U.S. Bureau of Land Management, 1958).[1] As such, those of Mexican descent comprised a majority of the Southwest population for much of the latter half of the 19th century. However, as Anglo-Americans migrated westward, Hispanic enclaves formed in north-central New Mexico and in southern regions of Arizona, Texas, and Colorado. Prior to the creation of the US Border Patrol in 1925, Mexicans immigrated with relative ease and thus were the primary source of immigrant farm labor. Approximately 700,000 Mexicans migrated to the United States in the first two decades of the 20th century, and most often located in newly irrigated agricultural regions in the West (Haverluk, 1997). The 1930 census counted over 1.5 million Hispanics in the US, 97 percent of whom were of Mexican origin[2], 92 percent were living west of the 100th meridian, and roughly a third were men between the ages of 20 and 50.

The Great Depression amplified hostilities towards Mexicans, and recent migrants were scapegoated for taking job opportunities away from the White population. Repatriation campaigns urging Mexicans to

[1] The treaty also included small pieces of land in present-day Wyoming and Kansas.

[2] The 1930 census was the first and only census to count Mexicans as a separate race. The remaining 3 percent were identified as Hispanic due to their Puerto Rican origin.

return across the border began in earnest in 1929 and persisted throughout much of the 1930s.[3] Due to the informal and localized nature of repatriation efforts, there is no formal documentation of the number of Mexicans expelled, but scholarly estimates range anywhere from 300,000 to over one million (Gratton and Merchant, 2013). In the Southwest, the population of first and second-generation Mexican migrants was nearly halved between 1930 and 1940 (see appendix figure 1).[4]

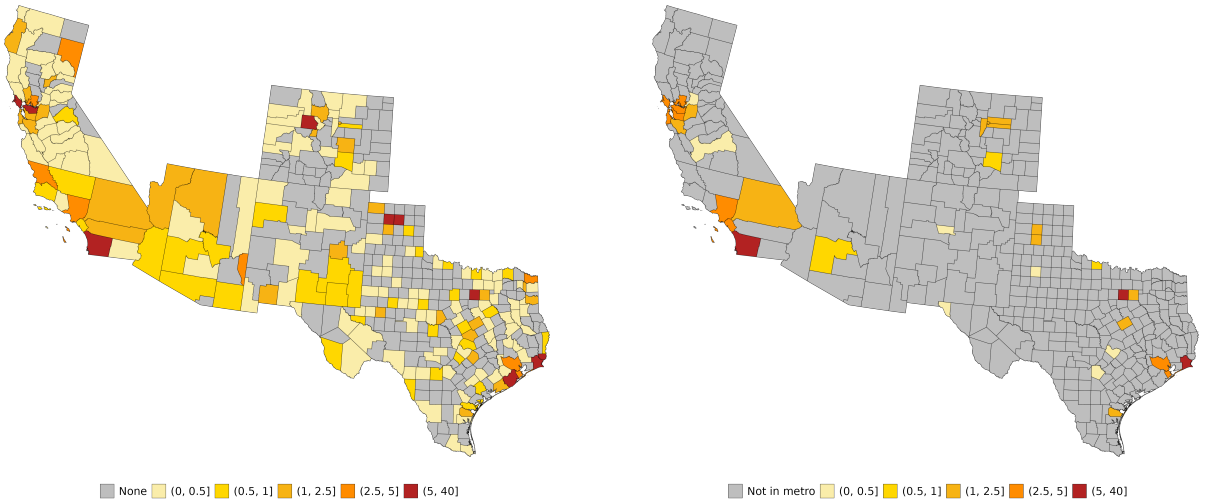
As of the 1940 census, approximately 400,000 Hispanic men were in the labor force in the Southwest. Of these, 37 percent were working in agriculture, predominantly as wage workers, and almost 20 percent were employed as non-farm laborers. The majority (54 percent) were living outside of metropolitan areas, and 16 percent were unemployed when the 1940 census was enumerated. The Hispanic unemployment rate was higher in metropolitan areas (17.7 percent versus 14.6 percent in non-metro areas), as was the average gap in annual earnings between Hispanic and non-Hispanic White men ($690 versus $511 (1940 dollars) in non-metro areas). While men employed in agriculture had the lowest annual earnings ($357 in 1940 dollars), the earnings gap between Hispanic and non-Hispanic white men was smallest in agriculture relative to all other occupation groups.

2.2 The Effect of WWII

WWII had several effects on the US economy and labor market. From 1940 to 1945, the federal government spent approximately $3.1 trillion (2014 $) on war-related production, not including manufacturing expenditures related to food processing, which comprised 40 percent of annual GDP from 1943 to 1945 (Aizer et al., 2022). Roughly 14 percent ($420.1 billion, 2014$) was spent in the Southwest, and the vast majority of Southwest expenditures (88 percent) went to 36 counties that comprised 25 metropolitan areas. On average, non-metro counties received $385 per capita of production expenditures (1940 $), while metro counties on average received $2,480 (1940$) per capita. Figure 1 shows the distribution of expenditures by county and metro areas. With the exception of the Dallas and Fort Worth metro areas in eastern Texas, the bulk of manufacturing expenditures went to coastal counties near ports. Contracts for military equipment accounted for 85 percent of WWII expenditures nationally and 70 percent of expenditures in the Southwest, and the remaining expenditures were for the construction of new war facilities.

[3] Though many Mexicans did voluntarily repatriate, it is commonly understood that the majority of returns to Mexico were the result of coerced deportations by local government authorities. See Gratton and Merchant (2013) for variety of sources verifying this.

[4] It is not clear whether the entirety of the decline of the Mexican population is explained by returns to Mexico, but it almost certainly explains a large fraction (Lee et al., 2022).



(a) By county

(b) By metro area

Figure 1: WWII manufacturing expenditures per capita ($1000), excl. food processing

It is estimated that 15.8 million men served in the US military during WWII, roughly equivalent to 40 percent of the male labor force in 1940 (Aizer et al., 2022). Approximately 61 percent were drafted while the remaining 39 percent were volunteers, and approximately 61 percent of all enlisted persons (both men and women) were employed in combat-facing positions.[5] Approximately 1.4 million people enlisted in the Southwest, and the casualty rate for Southwest enlistees was roughly 2 percent. To address rampant labor shortages during wartime, President Roosevelt issued Executive Order 8802 in 1941, which stated that companies receiving war contracts must not discriminate based on "race, creed, color, or national origin" for employment, and established the Fair Employment Practice Committee (FEPC) to facilitate the employment of minorities in defense industries and investigate discrimination disputes. Prior to the war, employment discrimination was rampant nationwide. A 1940 survey performed by the Bureau of Employment Security found that 51 percent of job openings in defense industries would not be made available to Black workers. In 1944, over 80 percent of FEPC complaints were for discrimination on the basis of race, 9 percent cited discrimination on the basis of religion (primarily against Jewish individuals), and 10 percent cited discrimination based on national origin (primarily against Mexican-Americans).[6] The FEPC was ef-

[5]These statistics are from the National WWII museum website at https://www.nationalww2museum.org/students-teachers/student-resources/research-starters/research-starters-us-military-numbers, and were accessed in February 2026.

[6]President's Committee on Fair Employment Practice. "First Report, July 1943-December 1944." Washington, D.C., 1945.

fective at increasing the share of non-White workers in defense industries in the short-run (Collins, 2001), however, because Mexicans are classified racially as White, the extent to which the executive order was effective at increasing Hispanic employment in defense industries is relatively unknown.

2.3 The Bracero Program

Farmers complained of significant farm labor shortages during WWII as men either enlisted in the armed forces or were pulled into jobs in defense industries. In order to alleviate these shortages, the US and Mexico negotiated a series of agreements that would allow farmers to import Mexican contract workers, or *braceros*, to fill seasonal farm labor gaps. Braceros were entitled to a number of job protections not afforded to other farm workers, including a government-set minimum wage, free housing and transportation, and guaranteed employment for at least 75 percent of the contract duration. Bracero contracts were only supposed to be approved if farmers could credily prove that there was an insufficient number of domestic workers willing and able to work at the bracero minimum wage, but it was not uncommon for braceros to be used as strikebreakers (Galarza, 1964). The initial agreement was part of the broader Emergency Farm Labor Supply Program (EFLSP) that was in effect from August 1942 until the end of 1947, during which the federal government subsidized the recruitment, importation, and placement of nearly 220,000 braceros (Martin, 2020).[7] The importation of braceros continued after the conclusion of the original agreement under the auspices of the 1917 Immigration Act until the program was formally reintroduced in 1951.

The relative appeal of US farm employment compared to employment opportunities in Mexico meant that the supply of bracero labor often far exceeded demand, and illegal immigrants surged into the Southwest. The ratio of illegal apprehensions by the Immigration and Naturalization Service (INS) to bracero contract workers was over 3 to 1 from 1942 to 1950, and over 6 to 1 from 1947 to 1950 (Martin, 2020). Migrating illegally allowed workers to avoid paying the bribes that were commonly expected to gain access to bracero recruitment centers, but precluded them from the mandated minimum wage and other protections afforded to braceros. Though it was illegal to knowingly house undocumented workers, it was not illegal to employ them, so farmers were relatively free to supplement their farm labor needs with undocumented workers when they were available. The intensity of immigration enforcement during the war was sporadic - border patrol agents were known to be acquiescent to demands from prominent agriculturalists belaboring la-

[7] The US also imported workers from Canada, Jamaica, Barbados, Honduras, and the Bahama Islands through the EFLSP, although employments of workers from these countries was minimal compared to bracero employment in the Southwest (Clemens et al., 2018).

bor shortages, particularly along the Texas border (Henderson, 2011). Mexico was able to leverage wartime labor shortages to demand stricter enforcement by threatening to withdraw from the Bracero agreement, but when the war ended and the US government stopped subsidizing the recruitment and transportation of braceros, farmers were even more inclined to employ illegal workers to avoid the administrative costs of the Bracero program and Mexico had little bargaining power to curtail it.

The US entering the Korean War in 1950 triggered a renegotiation of the Bracero agreement, which resulted in the Migrant Labor Agreement (MLA) of 1951. As part of the renegotiation, the population of undocumented workers currently employed on US farms were all given legal guest worker status, and immigration enforcement increased to discourage the use of illegal workers. The most famous effort to remove undocumented workers was Operation Wetback in 1954, when border patrol agents partnered with state and local police to remove over 1 million unauthorized Mexicans in the Southwest (García, 1980). Appendix figure 2 compares the number of bracero contracts to the number of "deportable aliens" apprehended by INS over the full duration of the program. Bracero contracts began to consistently exceed the number of apprehensions after Operation Wetback, although it is not clear to what extent this was driven by changes in enforcement. In general, utilization of the Bracero Program accelerated in the 1950s and reached a peak in 1956 when approximately 445,000 Mexicans were employed legally as braceros. At peak, braceros made up between 36 (California) and 87 (New Mexico) percent of the seasonal farm labor force in the Southwest (Clemens et al., 2018).

3 Data

My primary data source is full count census data from 1930, 1940, and 1950 from the Integrated Public Use Microdata Series (IPUMS) (Ruggles et al., 2025). In particular, I use the population linked across the 1940 and 1950 censuses by the Census Linking Project (Abramitzky et al., 2025, 2021), who use a New York State Identification and Intelligence System (NYSIIS) standardization algorithm to improve matches when names are potentially misspelled or mistranscribed. The NYSIIS algorithm is preferred in my setting as incidence of mistranscription is exacerbated for individuals with foreign names. The basic linking algorithm matches on first and last names, implied birth year (calculated from the census year and reported age), and place of birth, and observations with many possible matches are discarded. If there are no unique matches within the same birth year, the algorithm searches for unique name and birth place matches within 1 and

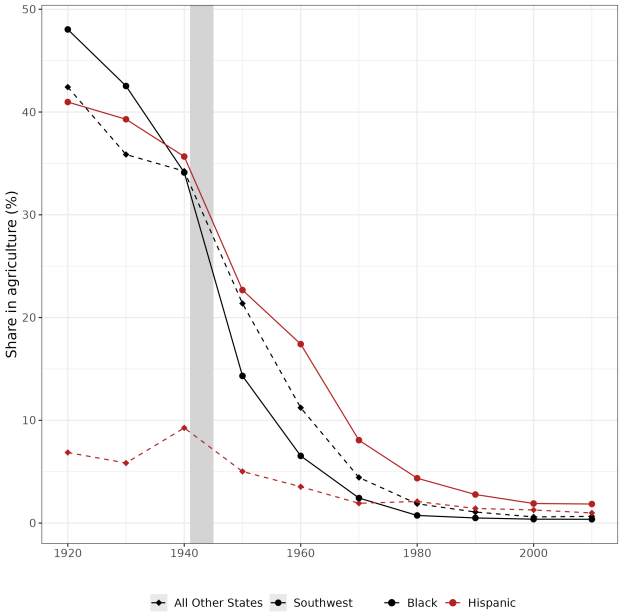
2 year bands acround a respondents' birth year. I additionally drop links where the race is inconsistent across the two census periods. Since the use of a Mexican race designation was discontinued after the 1930 census and individuals were not asked about Hispanic ethnicity until the 1980 census, I rely on the IPUMS classification of Hispanics in 1940. The majority (52 percent) of men classified by IPUMS as Hispanic in the linked sample were first or second-generation migrants from Hispanic countries (predominantly Mexico) and an additional 40 percent were classified as Hispanic by their surname. The remaining individuals were classified as Hispanic primarily based on the Hispanic status of their spouse or the household head (if related).

Data on WWII manufacturing expenditures by county primarily come from the 1947 County Data Book (CDB), which was compiled from the War Production Board's report on Major War Supply Contracts and Major War Facilities Projects from June 1940 through September 1945.[1] Contracts or projects valued over $50,000 in contemporary dollars are assigned to the county with the principal production plant, though projects related to food processing are excluded. Many of the Hispanic and Black men employed in the manufacturing industry were in occupations related to food processing, so I supplement these missing amounts with data digitized from a 1945 report from the War Production Board (WPB, 1945).[2] The 1947 CDB draws population, housing, and employment characteristics for all US counties from various Census data sources, and I also utilize the CDB's estimate of county population growth from 1940 to 1943. Farm sector data by county was obtained from the Census of Agriculture (COA).

Finally, I rely on recently digitized WWII enlistment and casualty records maintained by the National Archives and Records Administration that were made public by Ferrara (2024). The enlistment data contain rich demographic and geographic information for over 8 million men who were either drafted or enlisted in the armed forces between 1940 and 1945 and these data are then matched to a list of fallen soldiers in the Army and Air Force. However, the enlistments records do not include soldiers in other service branches like the Navy or Coast Guard, and some records are incomplete or otherwise messy due to scanning errors. To more closely approximate the impact that WWII service had on local labor markets, I supplement the enlistment records with data digitized from tables contained in a 1948 report from the Director of the Selective Service which detail the number of men inducted to the armed forces by service branch, month, and race (Selective Service System, 1948). These totals allow me to identify gaps in the enlistment data,

[1] The 1947 CDB is freely available through Haines et al. (2010).

[2] This report is available through HathiTrust and covers contracts for manufacturing facilities from June 1940 through August 1944.



Note: Solid lines correspond to trends in the Southwest and dashed lines correspond to all other states.

Figure 1: Share of male citizens employed in agriculture 1920-2010 by region

and I follow Aizer et al. (2022) by reweighting the observed enlistments to match the reported counts.[3]

3.1 Stylized Facts

Figure 1 shows long-run trends in agricultural employment for Black and Hispanic men both in and out of the Southwest. The share of Black men employed in agriculture in the Southwest was steadily falling before 1940, but the 59 percent decrease between 1940 and 1950 was the largest decline over one decade. The share of Hispanic men in agriculture fell by roughly a third between 1940 and 1950. Black men and Hispanic men in the Southwest in general held different agricultural occupations in 1940. 59 percent of Black men employed in agriculture were farmers or tenants, while nearly 73 percent of Hispanic men in agriculture were wage workers. The attachment of farmers and tenants during this period may be explained by unobserved characteristics, such as access to credit or tenancy contracts, so the primary focus of this paper is the relative attachment of Hispanics to farm wage work (FWW), which excludes farmers, tenants, and unpaid family workers.[4]

[3] If the aggregate number of Black enlistments to the Army in a given month is half of what is reported for that month in the Selective Service report, the observed individuals are given double the weight. This imposes the assumption that observable men have similar characteristics as the men not included in the enlistment data for a given race and month.

[4] The 1940 census only includes two FWW occupations - wage workers and farm foremen - while the 1950 census also includes farm services workers.

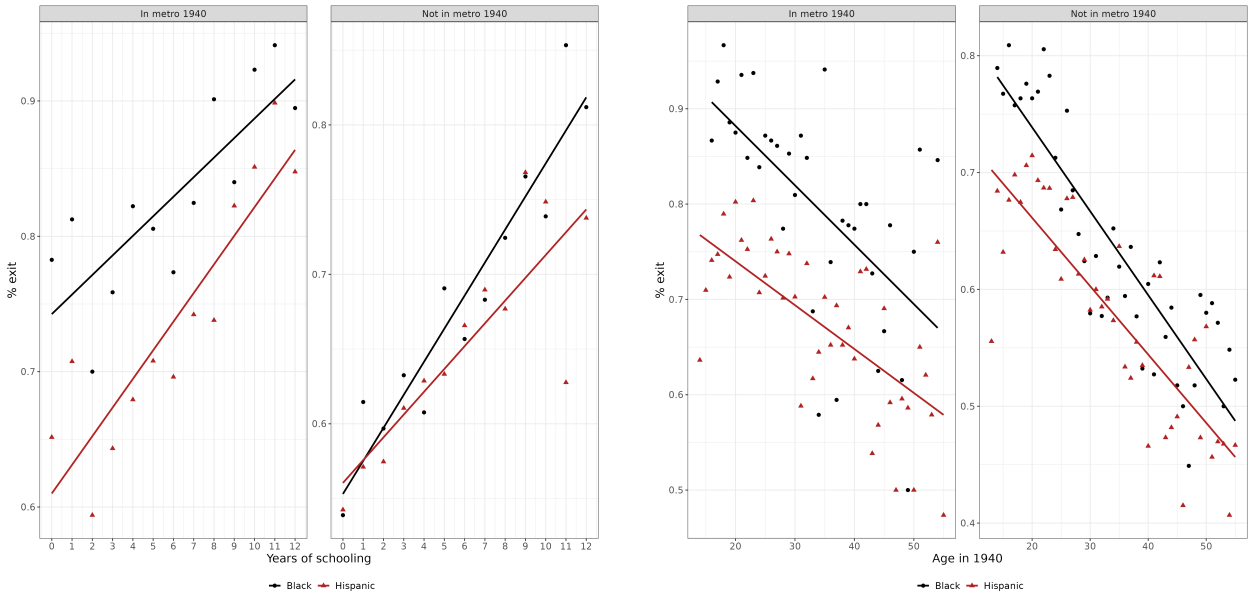
Table 1 shows the share of men who exited FWW occupations by farming region for the 10 regions with the largest number of farm workers in my sample. It is immediately clear that there was little overlap in where Hispanic and Black men worked as farm workers in 1940. Black men were concentrated in regions that specialized in field crops, particularly cotton and rice, while Hispanic men were more concentrated in regions producing specialty crops and livestock. There was significant variation in the introduction and adoption of agricultural technology over the 1940 decade across crops. In California, the share of cotton harvested mechanically rose by over 20 percentage points from 1940 to 1950 (U.S. Department of Agriculture, 1974), and the share of sugar beets harvested mechanically rose by almost 50 percentage points between 1945 and 1950 (Bailey, 1952). Adoption of mechanical harvesters for cotton was slower in Texas, where only 10 percent of the 1950 crop was mechanically harvested. The war also accelerated availability of chemical inputs, particularly nitrogen-based fertilizers, such that by 1954 approximately 58 percent of specialty crop acreage in the Southwest was treated with fertilizer (US Census Bureau, 1956).

		# in FWW occupations		Share exit	
Region	Primary type of farming	Black	Hispanic	Black	Hispanic
Black Waxy Prairies	Cotton	1,964	846	0.68	0.69
Southern California	Fruit, truck, and mixed	88	2,601	0.77	0.71
Lower Rio Grande Valley	Fruit, truck, and mixed	4	1,427	0.50	0.66
Western Sandy Coastal Plains	Cotton	1,321	4	0.69	0.75
Rio Grande Plains	Range livestock	5	1,198	0.80	0.62
San Joaquin Valley	Fruit, truck, and mixed	135	1,005	0.71	0.61
Gulf Coastal Prairies	Rice	679	260	0.75	0.69
Southern Arizona	Cotton, alfalfa	84	656	0.69	0.62
Post Oak Strip	Cotton, livestock	590	50	0.64	0.68
Central California Coast	Fruit, truck, and mixed	3	631	1	0.72
All Other	-	364	3,840	0.79	0.66

Note: The number employed is for men in my regression sample, which includes men in the linked population who were citizens, remained in the Southwest between census periods, and in the labor force in 1950.

Table 1: FWW employment in 1940 and exit shares by farm region

I find that exit rates from FWW occupations increased with education and decreased with age, but Black men were consistently more likely to exit at all levels of education. Figure 2 shows that this gap was most pronounced in metro areas, where the share of less-educated Black men exited FWW occupations at much higher rates than similarly-educated Hispanic men. Hispanic men were also more likely to enter FWW occupations than black men. Hispanic men living in metropolitan areas in 1940 with a grade-school



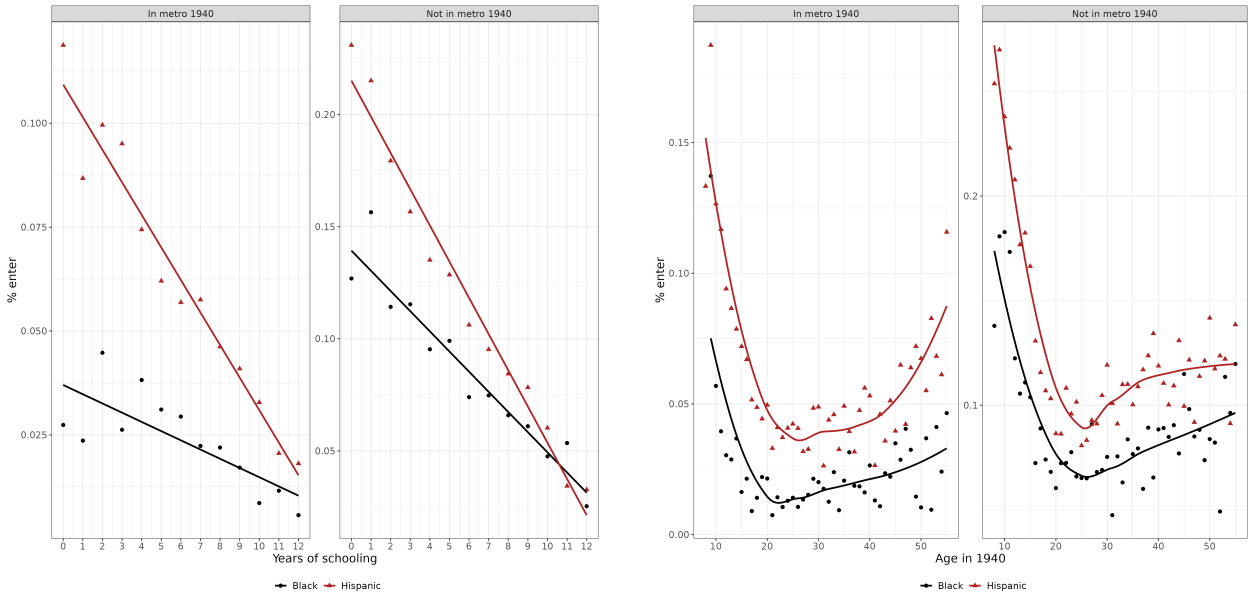
(a) Exit by years of schooling and metro status

(b) Exit by age and metro status

Note: Calculated using linked population from 1940 and 1950. Years of schooling is as of 1940, and I top code schooling at 12 years. Metropolitan status is assigned according to one's county

Figure 2: Exit from FWW occupations by education, age, and metropolitan status

education were nearly three times more likely to become wage workers than Black men, but this gap was smaller for those living in non-metro areas. Entry rates were also higher among Hispanics across all ages (see figure 3), and I note an interesting parabolic pattern for entry among both Hispanic and Black men. I do not currently have a hypothesis for why this pattern emerges, but it is something I plan to explore further.



(a) Entry by years of schooling and metro status

(b) Entry by age and metro status

Note: Calculated using linked population from 1940 and 1950. Years of schooling is as of 1940 for men who were not in school as of the 1940 census and as of 1950 for those who reported being in school in 1940. I top code schooling at 12 years.

Figure 3: Entry into FWW occupations by education, age, and metropolitan status

4 Empirical Approach

I first examine whether it is the case that observable characteristics explain the differential exit rates for Black and Hispanic men. I test this using a simple ordinary least squares (OLS) static regression on the sample of Black and Hispanic working men who were not in school in 1940 and reported working in FWW occupations, and who were in the labor force in 1950:

$$ExitAg_{ic} = \alpha + \beta_1 Hispanic_i + \beta_2 Niche_{ic}^{1940} + \boldsymbol{\beta}_3' \mathbf{X}_i + \boldsymbol{\beta}_4' \mathbf{Z}_c + \boldsymbol{\gamma}_r + \epsilon_{ic}, \tag{1}$$

where $ExitAg_{ic}$ is equal to 1 if individual i living in county c in 1940 was employed in a non-farm occupation in 1950, and $Hispanic_i$ is equal to 1 for Hispanic men and equal to 0 for Black men. To account for the influence of pre-existing coethnic networks in farm work, I include the binary variable $Niche_{ic}^{1940}$ that is equal to 1 if i's race/ethnicity group was over-represented in FWW occupations in county c in 1940. I follow Locke (2025) and define the representation index (RI) of group g in occupation o and county c as $RI_{goc} = \frac{P(g|o)_c}{P(g)_c}$, or the share of g employed in o in c given g's overall share of employment in c. I use his

threshold of 1.5, which implies that g was at least 50 percent over-represented in o, to determine the existence of *race/ethnicity occuaptional niches* in farm work. I also include a rich set of individual and county controls in $\mathbf{X_i}$ and $\mathbf{Z}_c$, respectively. The former includes a quadratic for age, years of schooling as of 1940, family size (binned, as of 1940), several dummy variables for 1940 characteristics (married, rented home, lived on farm, foreign-born), and an interaction for age and education.[1] The latter includes characteristics of the county where i lived in 1940: WWII manufacturing expenditures per capita (including contracts related to food processing), the draft rate,[2] the casualty rate of white soldiers from county c who were employed in semi-skilled occupations,[3] an estimate for population growth in county c between 1940 and 1943, and an indicator equal to 1 if county c was located in a metropolitan area. Finally, $\boldsymbol{\gamma}_r$ is a vector of 30 farm region dummies and ϵ_{ic} is the error term. Standard errors are clustered at the farm region level, which are groups of counties that are similar both in terms of geography and agricultural production.

In order to interpret β_1 as the Hispanic-specific effect on the probability of exiting FWW occupations, it must be the case that there are no omitted variables affecting one's exit decision. Unlike in previous censuses, individuals in 1940 and 1950 were not asked about English language ability (ELA), and this is likely to impact one's ability to get jobs in the non-farm sector. In lieu of self-reported ELA, I construct an ELA proxy using the share of individuals in each county that spoke English as of 1930 by race/ethnicity $\times$ nativity (REN) group (foreign-born Hispanic, US-born Hispanic, foreign-born White, US-born White, Southern-born Black, and non-Southern-born Black[4]) and occupation group[5]:

$$ELA_{goc} = \frac{\#\text{ of men of REN group } g \text{ in occupation group } o \text{ in county } c \text{ that spoke English in 1930}}{\#\text{ of men of REN group } g \text{ in occupation group } o \text{ in county } c \text{ in 1930}}. \quad (2)$$

I then assign ELA_{goc} to individuals in my data according to his county of residence in 1940, REN group, and occupation group.

I show the distribution of county proxy values for farm laborers by REN group in appendix figure 3. The share of Black and White farm workers who were English-speaking in 1930 is uniformly high. This is not

[1] I use years of schooling from the 1940 census because education information was only solicited from 20 percent of the population in 1950, and all the men in my sample reported being out of school in 1940.

[2] The draft rate is calculated as the number of men drafted from county c from 1940 to 1945 as a fraction of the working-age (18-65) male population in 1940. To account for migration that occurred during the war, I scale the denominator by an estimate of population growth from 1940 to 1943 (Aizer et al., 2022).

[3] This has shown to be predictive of entry into semi-skilled positions for Black workers in the South (Ferrara, 2022).

[4] Less than 1 percent of Black men in my sample were born outside of the US.

[5] I use IPUMS-defined occupation groups: professional and technical; farmers and farm managers; managers, officials, and proprietors (non-farm); clerical and kindred; sales workers; craftsmen; operatives; service workers; farm laborers; non-farm laborers; and unemployed. I also include an additional category for being in school.

the same for Hispanics; in the median county, only 51 percent of foreign-born Hispanic farm workers and 87 percent of US-born Hispanic farm workers reported speaking English in 1930. Within county $\times$ REN groups, there was also substantial variation in ELA across occupation groups for Hispanic men, particularly in large counties. In Los Angeles County, 95 percent of foreign-born Hispanic men employed as clerical or kindred workers reported speaking English in 1930, which is nearly double the share of those who were farm laborers (48 percent).[6] By relying on this 1930 measure, I am assuming that individuals were similarly sorted across occupations in that county in 1940. While I am unaware of any concrete evidence that this was not the case, it is possible that repatriation efforts in the 1930s weakened this relationship between ELA and occupation group within a county, either through selective migration or occupational downgrading of English-fluent Hispanics. Occupational downgrading would likely cause me to under-approximate ELA for certain subgroups, which could attenuate my estimates if the pattern was systematic. It is not obvious how selective migration between counties would impact my reliance on this proxy measure, but it is something that I plan to investigate further using the linked census population between 1930 and 1940.

One valid concern with using a county-level proxy is that the share of one's REN $\times$ occupation group that spoke English in 1930 may be correlated with other unobserved individual or county characteristics affecting one's decision to exit or enter the farm sector. I test whether my language proxy is predictive of county-level changes in the farm sector using data from the 1940 and 1950 COA. Specifically, I look at whether the share of Hispanic farm workers and the share of all Hispanics that spoke English in 1930 is predictive of changes in the number of farms, average farm size, average farm value, farmland share of county acres, and share of farms operated by tenants between 1940 and 1950, after controlling for other relevant county characteristics.[7] Results from this test are shown in appendix table 5, and reassuringly, they indicate that my ELA proxy for Hispanics is not predictive of county-level changes in the farm sector between 1940 and 1950. I have not yet determined the best way to test whether it is capturing other individual characteristics other than language because my proxy is constructed at the occupation-group level. Therefore, most of the outcomes I could realistically use as a falsification test with linked census data (e.g. change in marital status, family size, or county of residence) will likely be correlated with my proxy via occupation or in-

[6] Appendix figure 4 shows the distribution of ELA for Hispanic men in my regression samples by occupation group.

[7] I control for several 1940 county characteristics, including the log population, the employment share in agriculture and in manufacturing, share rural, share of farms with under $400 in sales, and the number of tractors. I also include controls to account for WWII, including the county draft rate, manufacturing expenditures per capita, and an estimate of population growth between 1940 and 1943.

come.[8] I recognize this as a shortcoming of my choice to use a proxy defined by occupation, and I welcome feedback on how to credibly test its validity.

My reliance on linked census data also makes my estimates susceptible to selection bias if the linked population is not representative of the full population (Ruggles, 2002). Appendix table 2 shows a balance table of demographic characteristics for the full and linked census population of Hispanic, Black, and White men in the labor force. Given that my sample sizes are rather large, I use a standardized mean difference statistic to test for meaningful differences in means separately by race/ethnicity group (Cohen, 2013), and I use a threshold of 0.1 standard deviations to determine statistical equivalence of two means. The balance test reveals that the linked population is actually rather similar to the full population on a number of characteristics, though in general linked men are younger, more educated, members of larger families, and less likely to have lived in group quarters. To improve the representativeness of the linked sample, I estimate propensity scores for inclusion in the linked sample using binned versions of the variables for which the linked population is meaningfully different from the full population (age, years of schooling, family size, living in group quarters), and incorporate inverse propensity score (IPS) reweighting as a robustness check (Bailey et al., 2020).

Finally, I also test whether Hispanic men were more likely to enter the farm sector between 1940 and 1950, I estimate equation 1 on the sample of Hispanic and Black men who were either in school or in another non-farm occupation in 1940 and I replace the outcome variable with $Enter_{ic}$ that is equal to 1 if individual i living in county c in 1940 was employed in a FWW occupation in 1950. I include the same controls as before, but for men who were in school in 1940 I use their reported years of schooling from 1950. I estimate all regressions with and without my language proxy, and I show estimates using IPS weights in the appendix. I drop men from my sample with undefined proxy values, which is the case if there were fewer than five men in one's REN $\times$ occupation group in that county in 1930. However, these men were mostly employed in more skilled occupations and thus presumably were less likely to transition into agriculture. Including these men in the regressions that exclude the ELA proxy does not change my entry results.

[8] Unsurprisingly, there is a marginally strong correlation between years of schooling and my ELA proxy among Hispanics in my sample ($\rho = 0.46$). However, it is relatively uncorrelated with the other demographic variables included in my regressions.

5 Main Results

Table 1 presents estimates for the parameters of interest in my regression on exit from FWW occupations, and the full results with and without IPS weights are shown in appendix table 3.[1] The first four columns are from my preferred specifiation with farm region fixed effects, and for comparison I show estimates using state fixed effects in columns (5) through (8). Model 1 includes only the Hispanic indicator and the relative fixed effects, model 2 adds the ELA proxy, model 3 adds the set of individual controls, and model 4 includes the full set of controls described in the previous section. I find no significant effect of being Hispanic on exit probabilities when I include farm region fixed effects. Importantly, I find that regional variation in agricultural production explains a lot about differential rates of exit between these two groups, as the Hispanic effect almost doubles when using state dummies instead of farm region dummies. This is consistent with the fact that Black men were concentrated in agricultural regions producing more readily mechanizable crops, so they were more free to enter the non-farm sector as adoption of harvesting technologies accelerated during the war.

Table 2 presents the coefficients of interest from my analysis on entry into FWW occupations, and the full results with and without IPS weights are shown in appendix table 4. I again present estimates obtained using farm region (columns 1-4) and state (columns 5-8) fixed effects, but in this case I do not find that regionality had much of an effect on entry behavior. In almost every specification, I find that the probability of entering FWW occupations was between 1 and 2 percentage points higher for Hispanic men than Black men. There was also a strong negative effect of ELA on entry probability. Holding everything else constant, moving from a proxy value of 0.5 to a value of 1 decreases the probability of entering the farm sector by 4 to 5 percentage points, and this effect is significant at the 1% level. The persistence of the Hispanic effect on entry patterns suggest that there was another Hispanic-specific force unrelated to language and geography that drew men into the farm sector.

6 Mechanisms

The results in the previous section indicate that observable characteristics do not explain why Hispanic men were more likely to enter FWW occupations over this period, so now I turn to further exploration of two

[1] Reassuringly, the weighted and unweighted estimates are highly similar.

	Dependent variable: Exited FWW occupations by 1950							
	Farm region dummies				State dummies			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Hispanic	-0.0338	0.0144	-0.0022	-0.0160	-0.0417**	0.0093	-0.0122	-0.0393
	(0.0252)	(0.0331)	(0.0360)	(0.0285)	(0.0180)	(0.0305)	(0.0410)	(0.0325)
ELA proxy		0.1386***	0.0348	0.0181		0.1042***	0.0375	0.0093
		(0.0227)	(0.0416)	(0.0409)		(0.0374)	(0.0668)	(0.0473)
Individual controls	No	No	Yes	Yes	No	No	Yes	Yes
County controls	No	No	No	Yes	No	No	No	Yes
IPS Weights	No	No	No	No	No	No	No	No
N Individuals	17,878	17,878	17,878	17,878	17,878	17,878	17,878	17,878
Mean Y	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
% Hispanic	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76
R^2	0.00877	0.01057	0.04942	0.05574	0.00288	0.00413	0.04433	0.05244
Within R^2	0.00038	0.00220	0.04137	0.04775	0.00129	0.00255	0.04280	0.05093

Note: Regression sample includes Hispanic and Black men who reported working as a farm wage worker or farm foreman in 1940 and were in the labor market in 1950. Clustered standard errors (farm region) shown in parentheses. *p<0.1; **p<0.05; ***p<0.01

Table 1: Coefficients of interest from equation 1

	Dependent variable: Entered FWW occupations by 1950							
	Farm region dummies				State dummies			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Hispanic	0.0368***	0.0193**	0.0127*	0.0105**	0.0488***	0.0218**	0.0190***	0.0143**
	(0.0049)	(0.0086)	(0.0064)	(0.0040)	(0.0097)	(0.0095)	(0.0062)	(0.0054)
ELA proxy		-0.1345***	-0.1006***	-0.0903***		-0.1517***	-0.1287***	-0.1112***
		(0.0234)	(0.0210)	(0.0211)		(0.0271)	(0.0217)	(0.0230)
Individual controls	No	No	Yes	Yes	No	No	Yes	Yes
County controls	No	No	No	Yes	No	No	No	Yes
IPS Weights	No	No	No	No	No	No	No	No
N Individuals	75,795	75,795	75,795	75,795	75,795	75,795	75,795	75,795
Mean Y	0.09	0.09	0.09	0.9	0.09	0.09	0.09	0.9
% Hispanic	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
R^2	0.02451	0.02921	0.06560	0.06899	0.00971	0.01625	0.05771	0.06282
Within R^2	0.00312	0.00793	0.04512	0.04858	0.00759	0.01415	0.05570	0.06082

Note: Regression sample includes Hispanic and Black men who were either in school or working in a non-farm job in 1940, and who were in the labor market in 1950. FWW occupations includes wage workers, farm foremen, and farm service workers. Clustered (farm region) standard errors are shown in parentheses. $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

Table 2: Coefficients of interest from entry analysis

potential mechanisms for the observed pattern. The first is that English deficiencies remained a barrier to entering the manufacturing sector, so low-ELA Hispanics were funnelled, or 'pushed', into the farm sector because they could not obtain manufacturing jobs. The second is that Hispanics were more complementary with new foreign farm workers from Mexico than non-Hispanic workers on account of shared language, so Hispanic workers were 'pulled' into farm work through increased demand. Since language barriers between workers can negatively impact productivity (Salleh et al., 2021; Johari and Jha, 2021), the test of the first hypothesis assumes that English language barriers were less prohibitive in areas where the network of Hispanics in manufacturing jobs was relatively large. For each county where at least 10 Hispanics were employed in 1940, I construct the representation index (RI) for Hispanics in manufacturing in 1940 using the same formula used to construct $Niche_{ic}$. I use IPUMS industry codes 306 through 499 to identify manufacturing workers, and I exclude people in occupations that are inconsistent with manufacturing (e.g. dancers, chiropractors). The RI is the ratio of the Hispanic share of manufacturing employment over the Hispanic share of the employed population in county c, and I say that Hispanics in county c were underrepresented in the manufacturing sector in 1940 if the RI is less than one. I then re-estimate my entry analysis on two subsamples, individuals living in 1940 counties where Hispanics were underrepresented (Under counties, $n = 200$) or proportionally represented (Over counties, $n = 158$) in the manufacturing sector in 1940. To account for the fact that the Hispanic manufacturing RI is likely inversely correlated with Hispanic networks in agriculture, I replace the binary $Niche_{ic}$ variable, which is equal to 1 if the RI for one's race/ethnicity group in FWW occupations exceeded 1.5, with the continuous measure of FWW RI by race/ethnicity and county of residence in 1940.

Since my ELA proxy is defined at the county $\times$ REN $\times$ occupation group level, one concern with this approach is that the Hispanic manufacturing RI in a county may be correlated with my ELA proxy in a way that makes these two groups of countys incomparable. I show county-level summary statistics for Over and Under counties in appendix table 7. Reassuringly, the average share of Hispanics who spoke English in 1930 is similar between these two groups of counties, with the share of Hispanic men in FWW occupations and overall being about 5 percentage points higher in Over counties. That being said, there are some noteworthy differences. Under counties were more rural, and a larger share of the employed population worked in agriculture. Under counties also received fewer WWII expenditures per capita, and perhaps consequently experienced more outmigration between 1940 and 1943. In general, counties with more war contracts experienced more inmigration between 1940 and 1943 while counties receiving fewer experienced more

outmigration, and the movement of new workers during the war likely affected the relative strength of Hispanic networks across space. Since migration between census periods may confound the results of this exercise, I recognize this as a shortcoming to my current approach, and I would appreciate feedback for alternative ways to test for the effect of language barriers for low-ELA Hispanics.

The results from this heterogeneity exercise are shown in table 1. For this specification, I weight individuals in my sample by the size of the employed Hispanic population (in log terms) in their 1940 county, because my RI measure is agnostic about population size. Two counties may have the same RI value and vastly different Hispanic population sizes, so this is intended to give more weight to individuals from counties with more employed Hispanics.[1] In general, the coefficients of interest are directionally similar across these county groups. I find that entry probabilities were 1.4 percentage points higher for Hispanic men relative to Black men in counties where Hispanics were proportionally represented in the manufacturing sector in 1940 (RI $\geq$ 1) even after accounting for ELA. Importantly, I find that the ELA effect is smaller in Over counties, which suggests that language was indeed a barrier to enter manufacturing jobs but that this effect was mitigated in areas with pre-existing coethnic networks. However, the fact that a Hispanic effect persists suggests that something other than language barriers drove entry decisions for Hispanics. Column 3 shows that ELA strongly predicted out-migration from these counties, such that a man with a proxy value of 0.5 was around 8 percentage points more likely to move out of an Over county than someone with high expected ELA. However, it seems that ELA did not affect entry decisions for the population of Hispanic men who migrated out of Over counties, as entry rates for this population ranged from 16 to 22 percent across the proxy distribution. Hispanic men who migrated out of Over counties were less educated than men who stayed regardless of ELA, so one explanation of this pattern is that manufacturing employment was only preferable over farm employment for higher-skilled men.

For men from counties where Hispanics were underrepresented in the manufacturing sector in 1940 (RI < 1), I find no Hispanic effect on entry into FWW occupations relative to Black men. However, there is a significant negative effect of ELA, suggesting that English proficiency expanded the set of non-farm opportunities regardless of networks. The results from column 4 indicate that Hispanic men were not less geographically mobile than Black men in these counties, which rules out the hypothesis that Black men were more willing to migrate to opportunity. I also find a negative relationship between ELA and probability of migrating out of Under counties, which is somewhat counterintuitive given that I would expect non-farm

[1] In general, this gives more weights to more populated counties, which also had more Black men.

opportunities to have been more limited in these counties. This may be due to the fact that there was more out-migration from these counties in general, which may have opened up manufacturing opportunities to high-ELA Hispanics.

	Dependent variable:			
	Entered FWW occupation		Moved counties	
RI	≥ 1	< 1	≥ 1	< 1
	(1)	(2)	(3)	(4)
Hispanic	0.0136***	0.0048	0.0203	0.0011
	(0.0038)	(0.0086)	(0.0268)	(0.0272)
ELA proxy	-0.0854***	-0.1227***	-0.1617***	-0.0578
	(0.0209)	(0.0347)	(0.0488)	(0.0469)
N Individuals	54,162	20,653	60,841	31,480
% Hispanic	0.67	0.68	0.69	0.70
Employment Weights	Yes	Yes	No	No
R^2	0.05348	0.07270	0.10871	0.06586
Adjusted R^2	0.05271	0.07063	0.10807	0.06450

Note: A representation index (RI) value of 1 means that the Hispanic share in manufacturing is proportional to their share of the population. FWW occupations includes wage workers, farm foremen, and farm service workers, and all regressions drop individuals who were farmers/tenants in either period. Robust standard errors shown in parentheses. *p<0.1; **p<0.05; ***p<0.01

Table 1: Heterogeneity by county Hispanic manufacturing RI in 1940

Since language and coethnic manufacturing networks do not fully explain Hispanic attachment to the farm sector, I now explore whether there was more demand for Hispanic farm workers given the influx of workers from Mexico. Specifically, I ask whether Hispanic men experienced more income gains from FWW incumbency than Black and White incumbents in 1950. I focus on incuments' income because many entrants had zero income in 1940, either from being unemployed or in school, so the population of men who *remain* in FWW occupations is rather different from the population of entrants. This also allows me to compare income growth of men for whom farm work was presumably their utility maximizing occupation. The reason I compare Hispanic incumbents to both Black and White incumbents is simply to increase my sample size, as the 1950 census only asked for income from 20 percent of the population.

I use Poisson quasi-maximum likelihood estimation (QMLE) instead of log-transforming my outcome because the former can accomodate zero-valued outcomes and will yield consistent estimates conditional

on correct specification of the conditional mean.[2] Another common way to deal with zero-valued outcomes is to use an inverse hyperbolic sine (or $arcsinh$) transformation, but Chen and Roth (2024) demonstrate that estimates obtained for an outcome variable transformed in this way are highly sensitive to the scale of the outcome variable, particularly when zeros are present.[3] Conversely, the signs of my Poisson QMLE estimates are stable across specifications and are therefore my preferred estimates. I estimate the following two-period difference-in-differences (DiD) model on the sample of Hispanic, Black, and White men who remained in FWW occupations across census periods:

$$Income_{irt} = \exp(\beta_1 Post_t + \beta_2 Hispanic_i \times Post_t + \boldsymbol{\beta}_3' \mathbf{X}_{it} + \boldsymbol{\gamma}_i + \boldsymbol{\gamma}_r)\epsilon_{irt}, \tag{1}$$

where $Income_{irt}$ is annual wage earnings (in 1999$) for individual i living in farm region r in year t, and $\mathbf{X}_{it}$ includes a quadratic for age, a marriage dummy, and a dummy for living in a metropolitan area. Finally, $\boldsymbol{\gamma}_i$ and $\boldsymbol{\gamma}_r$ are individual and farm region fixed effects respectively. I then estimate equation 1 with and without my ELA proxy interacted with $Post_t$ to assess the changing effect of ELA, while the level effect of ELA is absorbed by individual fixed effects. Here I use robust standard errors instead of clustering because the number of observations in each cluster are significantly reduced by data limitations, but I note that robust standard errors are likely deflated (Cameron and Miller, 2015).

A positive sign on β_2 would be consistent with increased complementarity of domestic Hispanics with the pool of new workers from Mexico. However, this interpretation is only valid under certain assumptions, namely that incomes for Hispanic, Black, and White farm workers in the Southwest were trending similarly prior to the policy change (in this case, the introduction of the Bracero Program), and would otherwise have continued trending similarly absent the policy change. Unfortunately, my setting precludes me from verifying the existence of parallel pre-trends as the census first asked respondents about income in 1940. Figure 1 shows that the share of Black men employed in agriculture in the Southwest was declining faster than that of Hispanic men in the decades leading up to 1940, which may indicate that farm incomes were declining for Black men. Though to the extent that there were regional differences in income trajectories, some of this effect will be absorbed by the farm region fixed effects.

The unweighted estimates for my parameters of interest are shown in table 2, and the full regresion re-

[2] Approximately 7 percent of regression observations have zero-valued income.

[3] Indeed, I ran an OLS version of equation 1 as a robustness check, and the estimates were very unstable depending on whether I transformed income in levels or in $1,000s.

sults with and without IPS weights are shown in appendix table 6. In the baseline model without accounting for ELA, Hispanic men experience significantly greater income growth between the two periods. The coefficient on $Post \times Hispanic$ suggests that income growth was 14 percent higher for Hispanic incumbents than Black and White incumbents, and this is significant at the 5% level. However, once the ELA proxy is included, the Hispanic effect shrinks and becomes statistically insignificant, suggesting that differences in English proficiency account for a substantial portion of the observed income growth differential. Columns (3) through (6) assess the sensitivity of these results to high-leverage income observations. In columns (3) and (4), the top 1 percent of income values are winsorized at the 99th percentile, and in columns (5) and (6) the top 5 percent are winsorized at the 95th percentile. I winsorize, or cap, high-leverage income observations to the 99th (95th) percentile values instead of omitting individuals from my regression to preserve as many data points as possible while still minimizing the influence of high-leverage observations. The coefficients in columns (3) and (4) are highly consistent with those obtained without winsorization, but winsorizing the top 5 percent of income values meaningfully attenuates my estimates of $\hat{\beta}_2$. The coefficient on $Post \times ELA$ remains negative and relatively large across all specifications, though becomes statistically insignificant when winsorizing the top 5 percent of income values.

	Dependent variable: Annual wage earnings							
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
Post	0.2398	0.6184	0.3865	0.7051*	0.3638	0.6075**	0.3828	0.7148*
	(0.3552)	(0.3896)	(0.3240)	(0.3761)	(0.2681)	(0.3056)	(0.3230)	(0.3754)
Post × Hispanic	0.1371**	0.0141	0.1365**	0.0318	0.0606	-0.0182	0.1377**	0.0293
	(0.0634)	(0.0901)	(0.0575)	(0.0810)	(0.0462)	(0.0653)	(0.0575)	(0.0807)
Post × ELA proxy		-0.3727*		-0.3165*		-0.2401		-0.3286*
		(0.2006)		(0.1915)		(0.1530)		(0.1903)
Farm region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Income winsorized	No	No	Top 99th	Top 99th	Top 95th	Top 95th	No	No
Diff. winsorized	No	No	No	No	No	No	1st/99th	1st/99th
N Individuals	2,356	2,356	2,356	2,356	2,356	2,356	2,356	2,356
% Hispanic	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
Squared Correlation	0.78364	0.78531	0.81067	0.81171	0.81345	0.81411	0.81144	0.81276
Pseudo R^2	0.73639	0.73694	0.74021	0.74061	0.72865	0.72893	0.74614	0.74658
BIC	8,968,700.7	8,950,043.4	8,513,716.5	8,500,427.7	7,274,034.0	7,266,684.8	8,396,903.0	8,382,539.9

Note: Robust standard errors shown in parentheses. Regressions include additional controls for age, marital status, and metro status. *p<0.1; **p<0.05; ***p<0.01

Table 2: Coefficients of interest from equation 1

The sensitivity of my results to high-valued observations requires further exploration. However, based on a preliminary investigation, I believe winsorizing the top 5 percent of income observations deflates the gains made by a subset of Hispanics at the low-end of the 1940 income distribution. Appendix figure 6 shows the coefficient estimates for the $Post_t \times Hispanic_i$ variable from my baseline regression without ELA by 1940 income quartile. Quartiles are defined based on the entire population of those in FWW occupations in 1940 who reported nonzero income in 1940. This exercise reveals the largest effect for men who were below the 25 percentile of income for FWW occupations in 1940, more moderate effects for those between the 25th and 50th percentile, and either null or negative effects for those in the top half of the distribution. However, the estimate for the first quartile, the magnitude of which implies a Hispanic effect of over 200 percent, strikes me as implausibly large. Inspection of the raw data leads me to believe this may be due to misreporting of annual income in one of the years, leading to extraordinarily large gains for a subset of observations. I plan to try alternative specifications winsorizing the *gains* in income, rather than the individual observations, to the 95th or 99th percentile to test for this pattern.

7 Discussion and Next Steps

In this paper, I have attempted to answer why Hispanic men in the Southwest states of California, Arizona, New Mexico, Colorado, and Texas were relatively more attached to farm work than Black men after WWII, in spite of the burgeoning manufacturing sector and influx of foreign farm workers from Mexico. My preliminary results suggest that this pattern was the culmination of three effects. The first is that since Black men were more often employed in the production of field crops in 1940, they were more exposed to the significant technological advancements for mechanical harvesting during and after the war that reduced labor demand. The second is that English deficiencies limited opportunities for Hispanics in the manufacturing sector. Finally, I have preliminary evidence that demand for Spanish-speaking farm workers increased due to their complementarity with the new supply of Mexican workers resulting from the Bracero program.

I currently have two directions that I hope to further extend this research, and I welcome any suggestions for additional directions to take this. The first is to assess the long-run impacts of remaining in agriculture on the economic prosperity of Hispanics in the Southwest. Though full census data are unavailable after 1950, there exists some county-level data on use of braceros after the program was re-established in 1951. Illegal migration from Mexico slowed considerably after Operation Wetback in 1953 and 1954 (Martin,

2020), so the maximum number of braceros employed in a county after 1954 should be a relatively close approximation for the number of foreign Mexican workers. Figure 1 shows a large decline in the share of Hispanics employed in agriculture between 1960 and 1970, which may in part be explained by the termination of the Bracero program in 1964. Therefore, I could then use variation in use of the Bracero program to approximate variation in demand for Spanish-speaking domestic workers to assess economic outcomes for Hispanics in 1960 and 1970.

Second, I hope to use a theoretical model to explore the role of coethnic networks in agriculture. Specifically, I want to ask how changes to the composition of farm workers from being predominantly White to predominantly Mexican affected the occupational choice of US-born men of Mexican ethnicity. I envision using a modified version of the occupational choice model from Roy (1951) that accounts for occupational networks and language as an occupation-specific skill. I can simulate agricultural exit and entry under the observed distribution of ELA, taking into account the degree of manufacturing expansions in one's geography. My empirical results suggest that language alone does not explain Hispanic attachment to the farm sector, so I would ideally like to use a dynamic model of occupational choice in order to assess whether there exists a turning point in agriculture when the size of one's coethnic network is more predictive of choosing that occupation than language.

Additional Tables and Figures

Chapter 1

			1950 Occupation				
	1940 Occupation	Total	F/Ts	Foremen/WWs	Other farm	Manuf.	Other non-farm
Black	F/Ts	8,303	0.43	0.09	0.01	0.09	0.38
Black	Foremen/WWs	5,721	0.13	0.15	0.01	0.12	0.59
Hispanic	F/Ts	6,496	0.34	0.14	0.01	0.05	0.46
Hispanic	Foremen/WWs	18,362	0.06	0.28	0.01	0.10	0.55

Note: Calculated using linked population from 1940 and 1950, with inverse propensity score weights (Bailey et al., 2020). WWs = wage workers, F/Ts = farmers/tenants. The number of Hispanic and Black foremen were relatively small - 164 and 11, respectively - so I grouped them with wage workers. The "other farm" category includes farm service workers and unpaid family workers.

Table 1: Exit by 1940 occupation

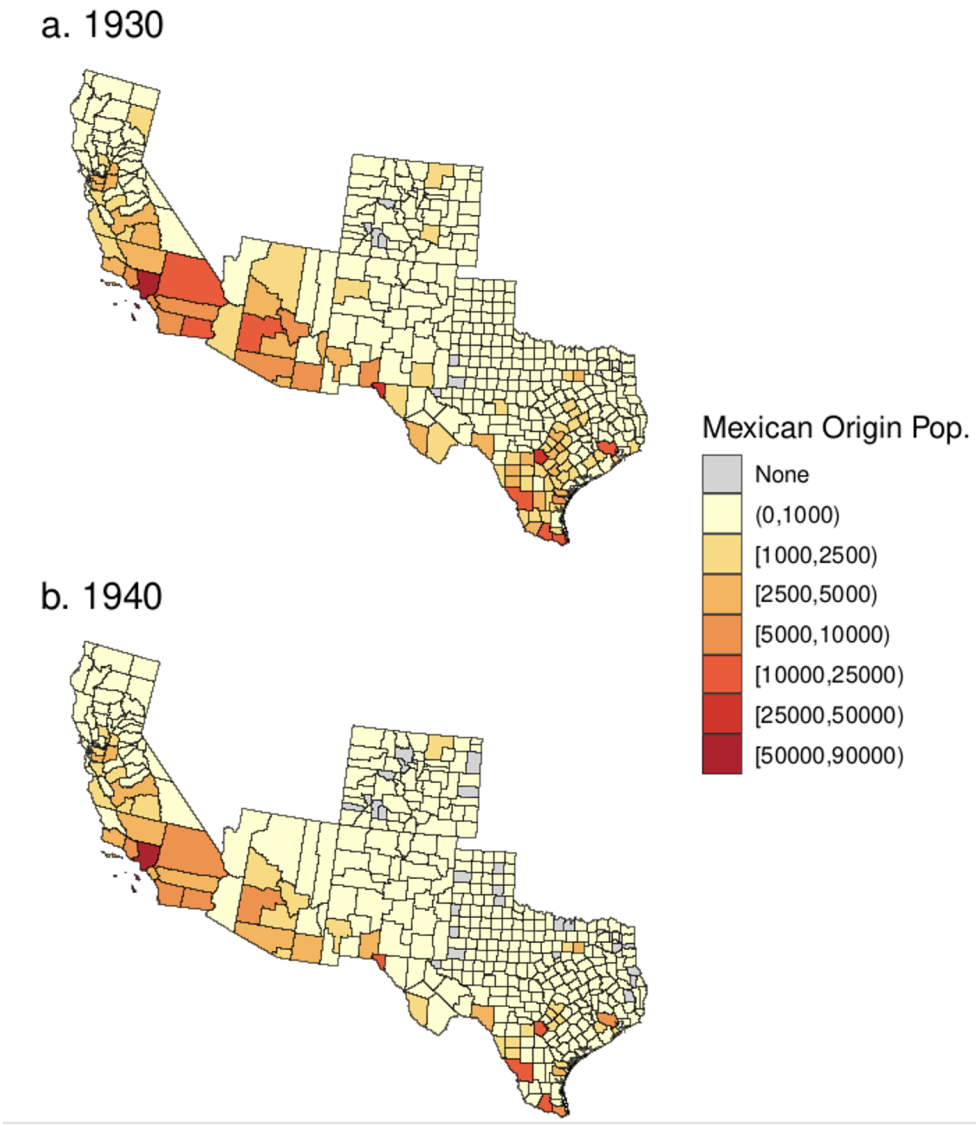


Figure 1: Mexican population, 1930 and 1940

Note: Map shows the population of first and second-generation Mexican men ages 18-65 enumerated in the 1930 and 1940 censuses.

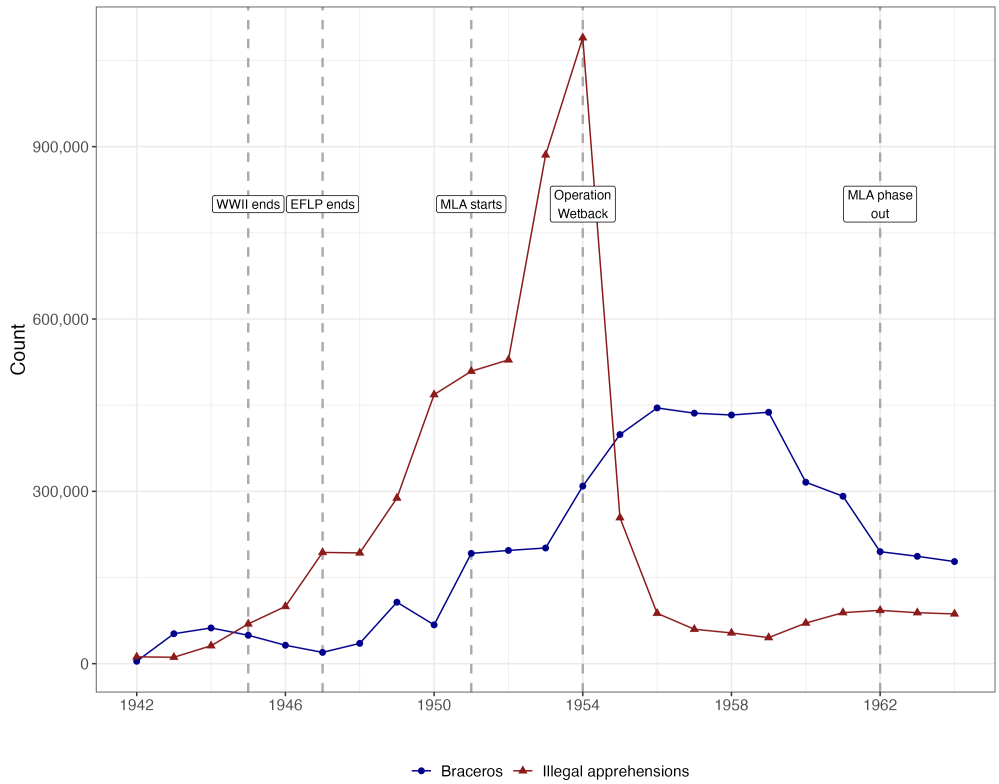
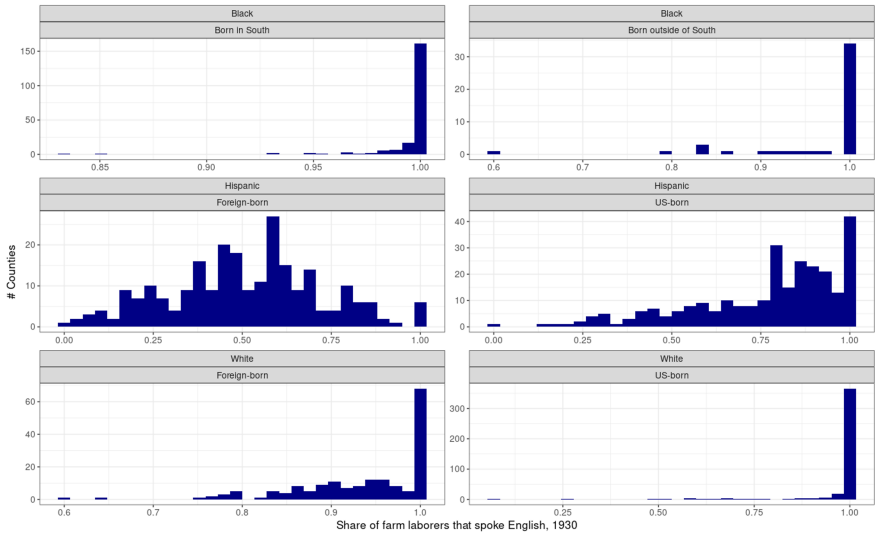


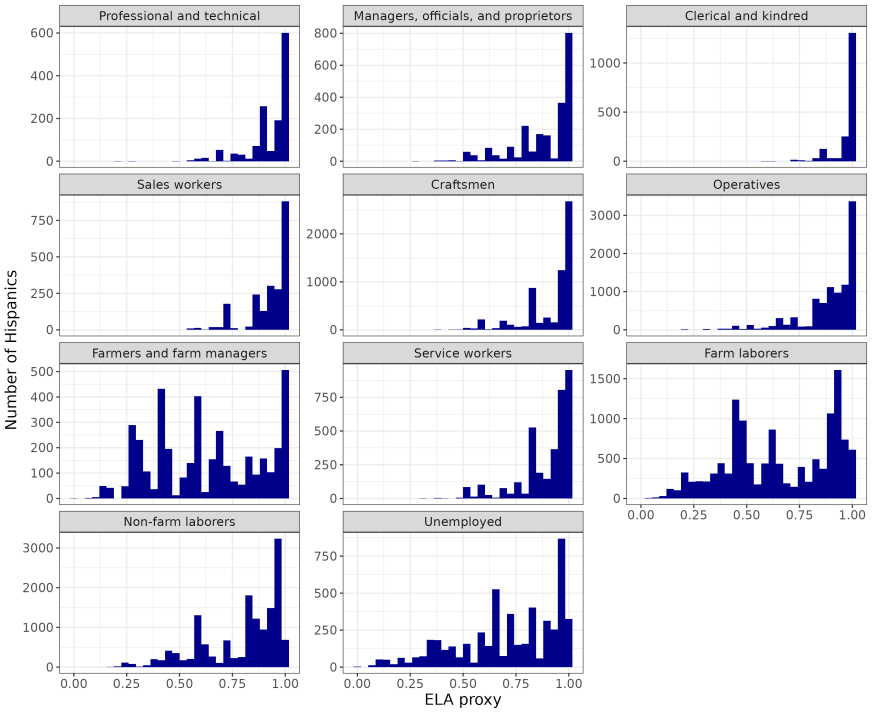
Figure 2: Bracero contracts and non-citizen apprehensions, 1942-1964

Source: Vialet and McClure (1980)



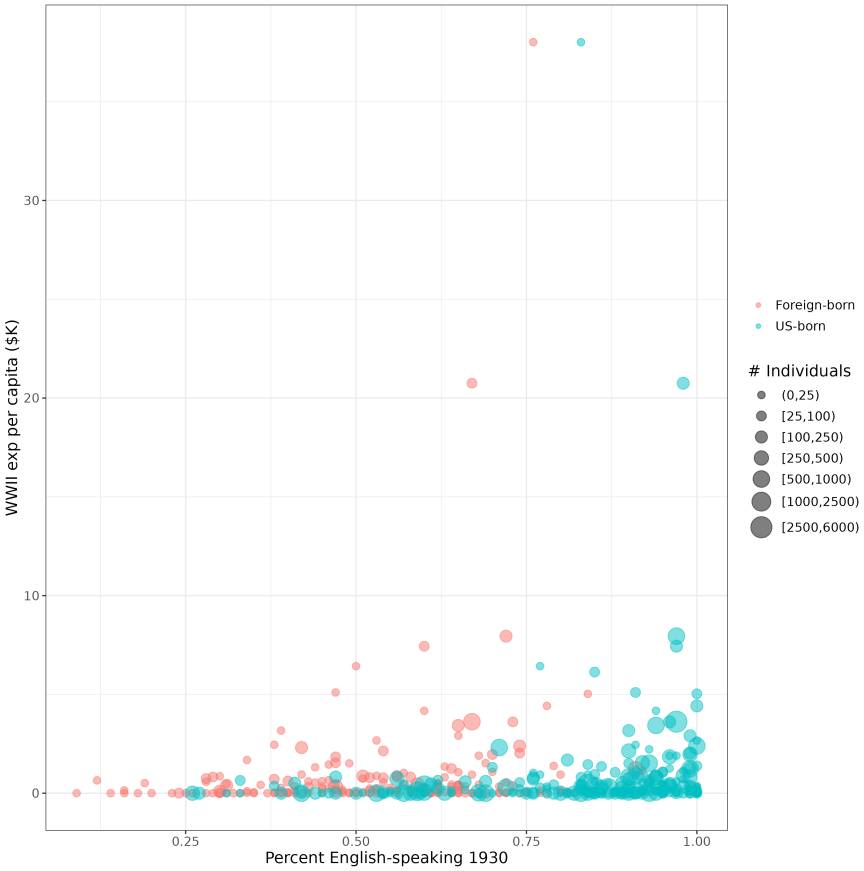
Note: Shares are computed for the working-age (18-65) population of male farm workers using the full count 1930 census data.

Figure 3: Distribution of ELA proxy by county and race/ethnicity × nativity (REN) group



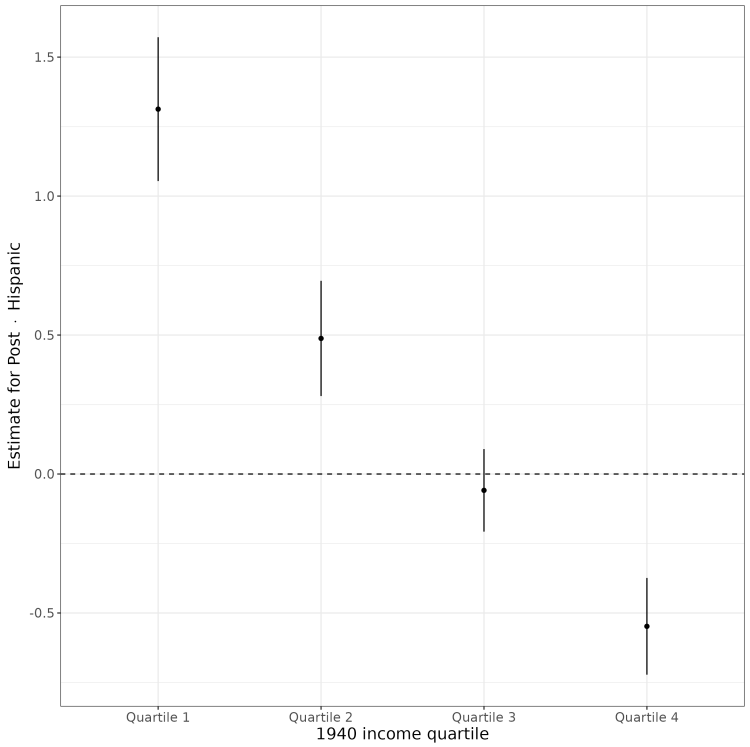
Note: ELA proxy distribution for Hispanic men in my exit and entry samples according to their occupation in 1940. The category for being in school is omitted as this applies to a small number of observations.

Figure 4: Distribution of ELA proxy for Hispanic men in regression samples



Note: The size of each dot corresponds to the number of Hispanic men in my sample, after applying inverse propensity score weights. Language proxy and war expenditures are shown for individuals' county in 1940.

Figure 5: Distribution of Hispanic men by per-capita war expenditures and language proxy



Note:

Figure 6: Estimates for $\hat{\beta}_2$ from equation 1 by 1940 income quartile

Variable	Full Mean (SD)	Linked Mean (SD)	Linked - Full	Std. Mean Diff.	SMD Desc.	N Full	Share Linked
			Hispanic Men				
Age	29.443 (14.613)	27.327 (12.377)	-2.116	-0.156	Small (<0.2)	582,591	0.23
Family size	5.737 (3.073)	5.839 (2.949)	0.102	0.034	Equiv. (<0.1)	582,591	0.23
Married	0.450 (0.497)	0.461 (0.499)	0.011	0.023	Equiv. (<0.1)	582,591	0.23
Renter	0.575 (0.494)	0.557 (0.497)	-0.018	-0.036	Equiv. (<0.1)	582,591	0.23
Has children	0.368 (0.482)	0.379 (0.485)	0.011	0.022	Equiv. (<0.1)	582,591	0.23
Years of schooling	5.114 (3.672)	5.822 (3.676)	0.708	0.193	Small (<0.2)	569,313	0.23
Parent foreign-born	0.035 (0.185)	0.034 (0.180)	-0.002	-0.009	Equiv. (<0.1)	582,591	0.23
Group quarters	0.032 (0.176)	0.022 (0.147)	-0.010	-0.062	Equiv. (<0.1)	582,591	0.23
			Black Men				
Age	31.931 (14.561)	29.212 (12.448)	-2.719	-0.201	Large (>=0.2)	407,782	0.18
Family size	4.211 (2.866)	4.624 (2.915)	0.413	0.143	Small (<0.2)	407,782	0.18
Married	0.530 (0.499)	0.542 (0.498)	0.011	0.023	Equiv. (<0.1)	407,782	0.18
Renter	0.648 (0.478)	0.642 (0.479)	-0.006	-0.012	Equiv. (<0.1)	407,782	0.18
Has children	0.276 (0.447)	0.307 (0.461)	0.031	0.068	Equiv. (<0.1)	407,782	0.18
Years of schooling	6.157 (3.407)	6.472 (3.355)	0.315	0.093	Equiv. (<0.1)	399,209	0.18
Parent foreign-born	0.000 (0.021)	0.000 (0.018)	-0.000	-0.006	Equiv. (<0.1)	407,782	0.18
Group quarters	0.042 (0.200)	0.030 (0.169)	-0.012	-0.066	Equiv. (<0.1)	407,782	0.18
			White Men				
Age	34.334 (15.076)	30.902 (12.468)	-3.433	-0.248	Large (>=0.2)	5,014,112	0.27
Family size	3.635 (2.045)	3.871 (1.971)	0.236	0.117	Small (<0.2)	5,014,112	0.27
Married	0.570 (0.495)	0.579 (0.494)	0.009	0.019	Equiv. (<0.1)	5,014,112	0.27
Renter	0.508 (0.500)	0.505 (0.500)	-0.003	-0.006	Equiv. (<0.1)	5,014,112	0.27
Has children	0.363 (0.481)	0.391 (0.488)	0.028	0.058	Equiv. (<0.1)	5,014,112	0.27
Years of schooling	9.301 (3.516)	9.613 (3.379)	0.312	0.090	Equiv. (<0.1)	4,927,484	0.27
Parent foreign-born	0.012 (0.108)	0.011 (0.107)	-0.000	-0.002	Equiv. (<0.1)	5,014,112	0.27
Group quarters	0.056 (0.229)	0.032 (0.177)	-0.023	-0.113	Small (<0.2)	5,014,112	0.27

Table 2: Standardized mean difference between Full and Linked census samples 1940-1950.

	Dependent variable: Exited FWW occupations by 1950							
	Unweighted				IPS weights			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Hispanic	-0.0338	0.0144	-0.0022	-0.0160	-0.0326	0.0109	-0.0040	-0.0190
	(0.0252)	(0.0331)	(0.0360)	(0.0285)	(0.0253)	(0.0329)	(0.0369)	(0.0294)
ELA proxy		0.1386***	0.0348	0.0181		0.1208***	0.0262	0.0039
		(0.0227)	(0.0416)	(0.0409)		(0.0250)	(0.0472)	(0.0486)
Foreign-born			0.0307	0.0226			0.0342	0.0235
			(0.0188)	(0.0188)			(0.0211)	(0.0222)
Age			-0.0061	-0.0061			-0.0075	-0.0076
			(0.0050)	(0.0050)			(0.0047)	(0.0048)
Age sq.			3.25×10^{-5}	3.14×10^{-5}			4.7×10^{-5}	4.66×10^{-5}
			(5.39×10^{-5})	(5.42×10^{-5})			(5.17×10^{-5})	(5.27×10^{-5})
Married			-0.0199**	-0.0192**			-0.0150*	-0.0139*
			(0.0076)	(0.0072)			(0.0085)	(0.0080)
Education			0.0280***	0.0270***			0.0283***	0.0272***
			(0.0063)	(0.0058)			(0.0063)	(0.0057)
Family size (binned)			-0.0248***	-0.0241***			-0.0269***	-0.0260***
			(0.0055)	(0.0055)			(0.0064)	(0.0064)
Farm status			-0.1099***	-0.1000***			-0.1095***	-0.0983***
			(0.0167)	(0.0149)			(0.0183)	(0.0162)
Renter			0.0047	0.0020			0.0043	0.0015
			(0.0081)	(0.0075)			(0.0095)	(0.0088)
Age x Education			-0.0003**	-0.0003**			-0.0004**	-0.0003**
			(0.0002)	(0.0001)			(0.0002)	(0.0001)
Metro				0.0656***				0.0694***
				(0.0151)				(0.0157)
In niche				-0.0047				-0.0041
				(0.0199)				(0.0214)
War exp per capita				0.0003				-0.0007
				(0.0059)				(0.0059)
Draft rate				-0.2660				-0.1981
				(0.1969)				(0.1860)
White SS casualty rate				0.4135*				0.4580**
				(0.2113)				(0.2211)
Pop. change 40-43				0.0962				0.1310
				(0.0828)				(0.0854)
Individual controls	No	No	Yes	Yes	No	No	Yes	Yes
County controls	No	No	No	Yes	No	No	No	Yes
IPS weights	No	No	No	No	Yes	Yes	Yes	Yes
N Individuals	17,893	17,893	17,893	17,893	17,893	17,893	17,893	17,893
R^2	0.00877	0.01057	0.04942	0.05574	0.00882	0.01019	0.04807	0.05529
Within R^2	0.00038	0.00220	0.04137	0.04775	0.00036	0.00175	0.03995	0.04723

Note: All regressions include farm region fixed effects. Refer to the text surrounding equation 1 for descriptions of other controls. Clustered (farm region) standard errors shown in parentheses. *p<0.1; **p<0.05; ***p<0.01

Table 3: Full regression results for equation 1

	Dependent variable: Entered FWW occupations by 1950							
	Unweighted				IPS Weights			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Hispanic	0.0368***	0.0193**	0.0127*	0.0105**	0.0376***	0.0183**	0.0143**	0.0126***
	(0.0049)	(0.0086)	(0.0064)	(0.0040)	(0.0048)	(0.0081)	(0.0069)	(0.0039)
ELA proxy		-0.1345***	-0.1006***	-0.0903***		-0.1379***	-0.1104***	-0.0987***
		(0.0234)	(0.0210)	(0.0211)		(0.0214)	(0.0214)	(0.0216)
Foreign-born			-0.0254***	-0.0222***			-0.0266***	-0.0228***
			(0.0050)	(0.0049)			(0.0046)	(0.0046)
Age			-0.0118***	-0.0116***			-0.0100***	-0.0098***
			(0.0015)	(0.0015)			(0.0013)	(0.0013)
Age sq.			0.0001***	0.0001***			9.98×10^{-5}***	9.84×10^{-5}***
			(1.4×10^{-5})	(1.43×10^{-5})			(1.21×10^{-5})	(1.23×10^{-5})
Married			-0.0032	-0.0033			-0.0090**	-0.0090**
			(0.0030)	(0.0029)			(0.0034)	(0.0033)
Education			-0.0150***	-0.0148***			-0.0134***	-0.0130***
			(0.0022)	(0.0022)			(0.0021)	(0.0022)
Family size (binned)			0.0022	0.0021			0.0001	8.64×10^{-5}
			(0.0016)	(0.0014)			(0.0020)	(0.0018)
Farm status			0.0941***	0.0857***			0.0991***	0.0908***
			(0.0167)	(0.0137)			(0.0185)	(0.0153)
Renter			0.0104***	0.0117***			0.0086***	0.0101***
			(0.0025)	(0.0023)			(0.0027)	(0.0023)
Age x Education			0.0002***	0.0002***			0.0001***	0.0001***
			(3.6×10^{-5})	(3.64×10^{-5})			(3.4×10^{-5})	(3.48×10^{-5})
Metro				-0.0376**				-0.0403***
				(0.0138)				(0.0132)
In niche				0.0096				0.0079
				(0.0071)				(0.0069)
War exp per capita				0.0003				0.0005
				(0.0022)				(0.0023)
Draft rate				-0.0213				-0.0468
				(0.0882)				(0.0935)
White SS casualty rate				0.0954				0.0940
				(0.2092)				(0.1839)
Pop. change 40-43				-0.0334				-0.0365
				(0.0306)				(0.0283)
Individual controls	No	No	Yes	Yes	No	No	Yes	Yes
County controls	No	No	No	Yes	No	No	No	Yes
IPS weights	No	No	No	No	Yes	Yes	Yes	Yes
N Individuals	75,795	75,795	75,795	75,795	75,795	75,795	75,795	75,795
R^2	0.02451	0.02921	0.06560	0.06899	0.02345	0.02841	0.06015	0.06372
Within R^2	0.00312	0.00793	0.04512	0.04858	0.00323	0.00829	0.04069	0.04433

Note: All regressions include farm region fixed effects. Refer to the text surrounding equation 1 for descriptions of other controls. Clustered (farm region) standard errors shown in parentheses. *p<0.1; **p<0.05; ***p<0.01

Table 4: Full results for entry into FWW occupations

	Dependent variable: % change 1940-1950				
	Number of farms	Share farmland	Avg. farm size	% tenancy	Avg. farm value
% Eng. speaking - Hisp. FWWs, 1930	−0.037	−0.314	−0.333	−0.119	0.035
	(0.080)	(0.218)	(0.570)	(0.133)	(0.917)
% Eng. speaking - All Hisp., 1930	−0.105	0.346	0.876	0.019	0.240
	(0.096)	(0.300)	(0.804)	(0.193)	(1.157)
% in manuf.	−0.004	−0.018**	0.045	0.015	0.066
	(0.003)	(0.007)	(0.040)	(0.012)	(0.054)
% in ag.	−0.001	−0.008**	−0.006	0.001	−0.003
	(0.001)	(0.003)	(0.008)	(0.002)	(0.012)
% rural	−0.00004	0.006**	0.006	−0.0005	−0.001
	(0.001)	(0.002)	(0.005)	(0.001)	(0.008)
Draft rate	−0.061	1.716***	0.206	−0.522	0.989
	(0.214)	(0.525)	(1.891)	(0.528)	(3.351)
% farms w/ sales < $400	−0.003***	0.007**	0.034***	0.003	0.065***
	(0.001)	(0.003)	(0.011)	(0.003)	(0.015)
# tractors	0.0001***	−0.0001	−0.0002	0.0001***	0.0003
	(0.00002)	(0.0001)	(0.0002)	(0.00003)	(0.0002)
WWII exp. per capita	−0.007	−0.049**	−0.124	−0.023	−0.175*
	(0.007)	(0.024)	(0.084)	(0.016)	(0.091)
Log pop.	−0.013	0.069	−0.062	−0.099**	−0.635***
	(0.017)	(0.056)	(0.152)	(0.047)	(0.218)
Pop. chang 40-43	0.193***	0.954***	1.005	0.051	1.484*
	(0.063)	(0.370)	(0.642)	(0.117)	(0.886)
Constant	0.197	−0.894	−0.407	0.624	6.223***
	(0.197)	(0.584)	(1.320)	(0.471)	(2.060)
# Counties	296	296	296	296	296
R^2	0.277	0.158	0.189	0.071	0.197
Adjusted R^2	0.249	0.125	0.157	0.035	0.166

Note: Data on outcome variables come from the 1940 and 1950 Census of Agriculture, data used to construct the language proxy come from 1930 full count population census, and data for all other controls come from the 1947 County Data Book (Haines et al., 2010). Regressions are run at the county level and counties with missing values are dropped. Robust standard errors are shown in parentheses. *p<0.1; **p<0.05; ***p<0.01)

Table 5: Language proxy placebo test: county farm sector outcomes

	Dependent variable: Annual wage earnings			
	(1)	(1)	(2)	(2)
Post × Hispanic	0.1371**	0.1377**	0.0141	0.0275
	(0.0634)	(0.0696)	(0.0901)	(0.1090)
Post × ELA proxy			-0.3727*	-0.3217
			(0.2006)	(0.2225)
Post	0.2398	0.0905	0.6184	0.4129
	(0.3552)	(0.4059)	(0.3896)	(0.4558)
Age	0.0959**	0.1064**	0.0971**	0.1078**
	(0.0392)	(0.0442)	(0.0389)	(0.0439)
Age square	-0.0007***	-0.0006***	-0.0007***	-0.0006***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Married	0.1638***	0.1563***	0.1658***	0.1568***
	(0.0611)	(0.0604)	(0.0606)	(0.0600)
In metro	-0.0606	-0.0893	-0.0599	-0.0885
	(0.0707)	(0.0729)	(0.0706)	(0.0727)
Farm region	Yes	Yes	Yes	Yes
Individual	Yes	Yes	Yes	Yes
Winsorized	No	No	No	No
IPS Weights	No	Yes	No	Yes
N Individuals	2,356	2,356	2,356	2,356
% Hispanic	0.27	0.27	0.27	0.27
Squared Correlation	0.78364	0.78401	0.78531	0.78541
Pseudo R^2	0.73639	1.1908	0.73694	1.1905
BIC	8,968,700.7	10,359,066.7	8,950,043.4	10,342,467.3

Note: Robust standard errors shown in parentheses. *p<0.1; **p<0.05; ***p<0.01

Table 6: Full regression results for equation 1

Variable	RI < 1	RI ≥ 1
Hispanic English fluency (1930)	0.639	0.690
	(0.198)	(0.149)
Hispanic FWW English fluency (1930)	0.617	0.671
	(0.222)	(0.209)
Log population (1940)	9.607	9.854
	(1.015)	(1.321)
% change population (1940-1943)	-0.093	-0.029
	(0.201)	(0.294)
Share rural (1940)	0.769	0.682
	(0.22)	(0.278)
Male employment rate (1940)	0.798	0.795
	(0.041)	(0.044)
Share H.S. graduate (1940)	0.229	0.240
	(0.07)	(0.084)
Share employed in ag. (1940)	0.419	0.343
	(0.183)	(0.208)
Share employed in manuf. (1940)	0.053	0.08
	(0.056)	(0.088)
WWII exp. per capita (1940-1945)	0.384	0.918
	(0.991)	(3.593)

Table 7: County summary statistics by Hispanic 1940 manufacturing representation index (RI)

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